Wipro Capstone Project 3

Ashutosh Das (2141004162)

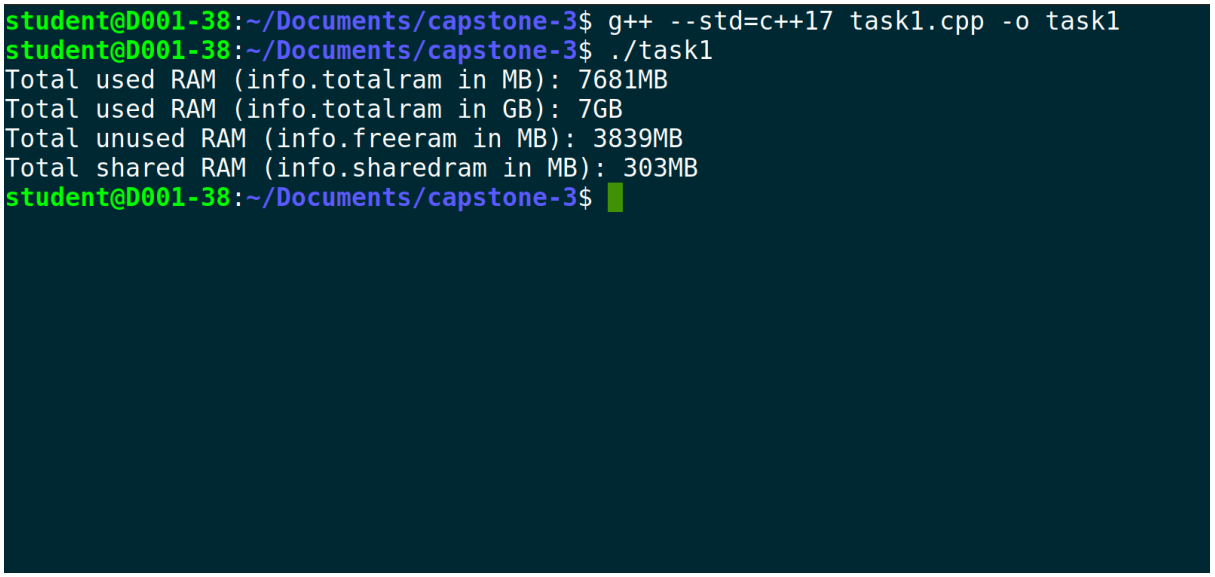
Objective: Create a system monitor tool in C++ that displays real-time information about system processes, memory usage, and CPU load, similar to the 'top' command.

Day-wise Tasks:

Day 1: Design UI layout and gather system data using system calls.

```
#include <iostream>

using namespace std;

#include <sys/sysinfo.h>

void displayMemoryInfo(){

	struct sysinfo info;

	if (sysinfo(&info)==0){

		cout<< "Total used RAM (info.totalram in MB): "<<info.totalram/(1024*1024)<<"MB\n";

		cout<< "Total used RAM (info.totalram in GB): "<<(info.totalram/(1024*1024))/1024<<"GB\n";

		cout<< "Total unused RAM (info.freeram in MB): "<<info.freeram/(1024*1024)<<"MB\n";

		cout<< "Total shared RAM (info.sharedram in MB): "<<info.sharedram/(1024*1024)<<"MB\n";

	}

}

int main(){

	displayMemoryInfo();

	return 0;

}
```

Day 1 Output



Day 2: Display process list with CPU and memory usage.

Process List Code

```
#include <iostream>
#include <filesystem>
#include <fstream>
#include <sstream>
#include <algorithm>
using namespace std;
namespace fs = std::filesystem;
bool isNumber(const string &s){
        return  !s.empty() && all_of(s.begin(), s.end(), ::isdigit);
}

string getProcessName(int pid){
        ifstream file("/proc/" + to_string(pid) + "/stat");
        string line, processName;
        if(file.is_open()){
                getline(file,line);
                istringstream ss(line);
```

```
            string token;
            int count=0;
            while(ss >> token){
                count++;
                if(count==2){
                    processName=token;
                    break;
                }
            }
        }
        return processName;
}

int main(){
        cout<<"Active processes: "<<endl;
        for (const auto &entry : fs::directory_iterator("/proc")){
            if(entry.is_directory()){
                string filename=entry.path().filename().string();
                if (isNumber(filename)){
                    int pid=stoi(filename);
                    string processName = getProcessName(pid);
                    cout<<"PID: "<<pid<<"|Name: " << processName<<endl;
                }
            }
        }
        return 0;
}
```

CPU and Memory Usage Code

```
#include <iostream>
#include <fstream>
#include <sstream>
#include <thread>
#include <chrono>
using namespace std;
struct CPUData{
	long user, nice, system, idle, iowait, irq, softirq, steal, guest, guest_nice;
};


CPUData getCPUData(){
	ifstream file("/proc/stat");
	string line;
	CPUData cpu={};
	if (file.is_open()){
		getline(file,line);
		istringstream ss(line);
		string cpuLabel;
		ss >> cpuLabel >> cpu.user >> cpu.nice >> cpu.system >> cpu.idle >> cpu.iowait >> cpu.irq >> cpu.softirq >> cpu.steal >> cpu.guest >> cpu.guest_nice;
	}
	return cpu;
}
double calculateCPUUsage (CPUData prev, CPUData current){
	long prevIdle= prev.idle + prev.iowait;
	long currIdle= current.idle + current.iowait;
	long prevTotal = prev.user + prev.nice + prev.system + prev.idle + prev.iowait + prev.irq + prev.softirq + prev.steal;
```

```
        long currTotal = current.user + current.nice + current.system + current.idle +
current.iowait + current.irq + current.softirq + current.steal;

        long totalDiff = currTotal - prevTotal;

        long idleDiff = currIdle - prevIdle;

        return ((totalDiff - idleDiff) * 100.0) / totalDiff;

}


int main(){

        CPUData cpu= getCPUData();

        cout << "User: " << cpu.user << endl;

    cout << "Nice: " << cpu.nice << endl;

    cout << "System: " << cpu.system << endl;

    cout << "Idle: " << cpu.idle << endl;

    cout << "IOwait: " << cpu.iowait << endl;

    cout << "IRQ: " << cpu.irq << endl;

    cout << "SoftIRQ: " << cpu.softirq << endl;

    cout << "Steal: " << cpu.steal << endl;

    cout << "Guest: " << cpu.guest << endl;

    cout << "Guest Nice: " << cpu.guest_nice << endl;

    CPUData prevData = getCPUData();

    this_thread::sleep_for(chrono::seconds(1));

    CPUData currData = getCPUData();

    double cpuUsage= calculateCPUUsage (prevData, currData);

    cout << "CPU Usage: "<<cpuUsage<<endl;

        return 0;

}
```

Day 2 Output (Process List)

```
student@D001-38:~/Documents/capstone-3$ g++ --std=c++17 task2_2.cpp -o task2_2
student@D001-38:~/Documents/capstone-3$ ./task2_2
Active processes:
PID: 1|Name: (systemd)
PID: 2|Name: (kthreadd)
PID: 3|Name: (rcu_gp)
PID: 4|Name: (rcu_par_gp)
PID: 5|Name: (slub_flushwq)
PID: 6|Name: (netns)
PID: 8|Name: (kworker/0:0H-events_highpri)
PID: 10|Name: (mm_percpu_wq)
PID: 11|Name: (rcu_tasks_rude_)
PID: 12|Name: (rcu_tasks_trace)
PID: 13|Name: (ksoftirqd/0)
PID: 14|Name: (rcu_sched)
PID: 15|Name: (migration/0)
PID: 16|Name: (idle_inject/0)
PID: 18|Name: (cpuhp/0)
PID: 19|Name: (cpuhp/1)
PID: 20|Name: (idle_inject/1)
PID: 21|Name: (migration/1)
PID: 22|Name: (ksoftirqd/1)
PID: 23|Name: (kworker/1:0-cgroup_destroy)
```

Day 2 Output (CPU and Memory Usage)

```
student@D001-38:~/Documents/capstone-3$ g++ --std=c++17 task2.cpp -o task2
student@D001-38:~/Documents/capstone-3$ ./task2
User: 155898
Nice: 574
System: 34903
Idle: 21088283
IOwait: 11435
IRQ: 0
SoftIRQ: 835
Steal: 0
Guest: 0
Guest Nice: 0
CPU Usage: 0.333333
student@D001-38:~/Documents/capstone-3$
```

Day 3: Implement process sorting by CPU and memory usage.

```
#include <iostream>
#include <filesystem>
#include <fstream>
#include <sstream>
#include <algorithm>
#include <vector>
#include <dirent.h>
#include <unistd.h>
#include <csignal>
using namespace std;
namespace fs=std::filesystem;
struct ProcessInfo{
        int pid;
        string name;
        double cpuUsage;
        long memoryUsage;
};
string readFileValue(const string &path){
        ifstream file(path);
        string value;
        if (file.is_open()){
                getline(file,value);
        }
        return value;
}
double getSystemUptime(){
        ifstream file("/proc/uptime");
        double uptime;
        if (file.is_open()){
```

```
            file >> uptime;
        }
        return uptime;
}
ProcessInfo getProcessInfo(int pid,double systemUptime){
        ProcessInfo pinfo;
    pinfo.pid = pid;
        ifstream file("/proc/" + to_string(pid) + "/stat");
        string line;
        long utime=0, stime=0, starttime=0;
        if(file.is_open()){
                getline(file,line);
                istringstream ss(line);
                string token;
                int count=0;

                while(ss >> token){
                        count++;
                        if(count==2) pinfo.name=token;
                        else if(count==14) utime=stol(token);
                        else if(count==15) stime=stol(token);
                        else if(count==22) starttime=stol(token);
                }
        }
        ifstream memFile("/proc/" + to_string(pid) + "/status");
        if (memFile.is_open()){
                string key, value,unit;
                while(memFile >> key >> value >> unit){
                        if (key=="VmRSS"){
                                pinfo.memoryUsage=stol(value);
                                break;
```

```
                }
            }
        }
        long total_time = utime + stime;
        double seconds = systemUptime - (starttime / sysconf(_SC_CLK_TCK));
        pinfo.cpuUsage = (total_time / sysconf(_SC_CLK_TCK)) / seconds * 100;
        return pinfo;
}
vector <ProcessInfo> getAllProcesses(){
        vector <ProcessInfo> processes;
        double systemUptime= getSystemUptime();
        for (const auto &entry : fs::directory_iterator("/proc")){
            if (entry.is_directory()){
                string filename=entry.path().filename().string();
                if (all_of(filename.begin(),filename.end(), ::isdigit)){
                    int pid=stoi(filename);

processes.push_back(getProcessInfo(pid,systemUptime));
                }
            }
        }
        return processes;
}
void sortProcesses(vector<ProcessInfo> &processes, bool sortByCpu){
        if(sortByCpu){
            sort(processes.begin(),processes.end(),[](const ProcessInfo &a, const
ProcessInfo &b)
            {
                return a.cpuUsage > b.cpuUsage;
            });
        }
```

```
        else{
                sort(processes.begin(),processes.end(),[](const ProcessInfo &a, const
ProcessInfo &b)
                {
                        return a.memoryUsage > b.memoryUsage;
                });
        }
}
int main(){
        vector <ProcessInfo> processes = getAllProcesses();
        sortProcesses(processes,true);
        cout<<"PID\tCPU%\tMemory (kB)\tName\n";
        for (size_t i=0;i<min(processes.size(),size_t(10));i++){

cout<<processes[i].pid<<"\t"<<processes[i].cpuUsage<<"%\t"<<processes[i].memory
Usage<<"%\t"<<processes[i].name<<"%\n";
        }
        return 0;
}
```

Day 3 Output

```
student@D001-38:~/Documents/capstone-3$ g++ --std=c++17 task3.cpp -o task3
student@D001-38:~/Documents/capstone-3$ ./task3
PID     CPU%    Memory (kB)     Name
3337    14.162% 140731003253864%        (chrome)%
1615    2.83279%        140731003253864%        (gnome-shell)%
2118    1.78108%        140731003253864%        (chrome)%
2077    0.937703%       140731003253864%        (chrome)%
1050    0.91241%        140731003253864%        (mysqld)%
1486    0.875344%       140731003253864%        (Xorg)%
4148    0.626913%       140731003253864%        (nautilus)%
2390    0.621781%       140731003253864%        (chrome)%
4076    0.337124%       140731003253864%        (gedit)%
3795    0.225338%       140731003253864%        (kworker/10:0-events)%
student@D001-38:~/Documents/capstone-3$
```

Day 4: Add functionality to kill processes.

```
#include <iostream>
#include <filesystem>
#include <fstream>
#include <sstream>
#include <algorithm>
#include <vector>
#include <dirent.h>
#include <unistd.h>
#include <csignal>
using namespace std;
namespace fs=std::filesystem;
struct ProcessInfo{
	int pid;
	string name;
	double cpuUsage;
	long memoryUsage;
};
string readFileValue(const string &path){
	ifstream file(path);
	string value;
	if (file.is_open()){
		getline(file,value);
	}
	return value;
}
double getSystemUptime(){
	ifstream file("/proc/uptime");
```

```
        double uptime;
        if (file.is_open()){
                file >> uptime;
        }
        return uptime;
}
ProcessInfo getProcessInfo(int pid,double systemUptime){
        ProcessInfo pinfo;
    pinfo.pid = pid;
        ifstream file("/proc/" + to_string(pid) + "/stat");
        string line;
        long utime=0, stime=0, starttime=0;
        if(file.is_open()){
                getline(file,line);
                istringstream ss(line);
                string token;
                int count=0;

                while(ss >> token){
                        count++;
                        if(count==2) pinfo.name=token;
                        else if(count==14) utime=stol(token);
                        else if(count==15) stime=stol(token);
                        else if(count==22) starttime=stol(token);
                }
        }

        ifstream memFile("/proc/" + to_string(pid) + "/status");
```

```
        if (memFile.is_open()){
                string key, value,unit;
                while(memFile >> key >> value >> unit){
                        if (key=="VmRSS"){
                                pinfo.memoryUsage=stol(value);
                                break;
                        }
                }
        }
        long total_time = utime + stime;
        double seconds = systemUptime - (starttime / sysconf(_SC_CLK_TCK));
        pinfo.cpuUsage = (total_time / sysconf(_SC_CLK_TCK)) / seconds * 100;
        return pinfo;


}

vector <ProcessInfo> getAllProcesses(){
        vector <ProcessInfo> processes;
        double systemUptime= getSystemUptime();
        for (const auto &entry : fs::directory_iterator("/proc")){
                if (entry.is_directory()){
                        string filename=entry.path().filename().string();
                        if (all_of(filename.begin(),filename.end(), ::isdigit)){
                                int pid=stoi(filename);

processes.push_back(getProcessInfo(pid,systemUptime));
                        }
```

```
            }
        }
        return processes;
}

void sortProcesses(vector<ProcessInfo> &processes, bool sortByCpu){
        if(sortByCpu){
                sort(processes.begin(),processes.end(),[](const ProcessInfo &a, const
ProcessInfo &b)
                {
                        return a.cpuUsage > b.cpuUsage;
                });
        }
        else{
                sort(processes.begin(),processes.end(),[](const ProcessInfo &a, const
ProcessInfo &b)
                {
                        return a.memoryUsage > b.memoryUsage;
                });
        }
}


int main(){
        vector <ProcessInfo> processes = getAllProcesses();
        sortProcesses(processes,true);
        cout<<"PID\tCPU%\tMemory (kB)\tName\n";
        for (size_t i=0;i<min(processes.size(),size_t(10));i++){
```

```
cout<<processes[i].pid<<"\t"<<processes[i].cpuUsage<<"%\t"<<processes[i].memory
Usage<<"%\t"<<processes[i].name<<"%\n";
        }
        int targetPid;
        cout<<"Enter PID to kill: ";
        cin>>targetPid;
        if(targetPid > 0){
                if (kill(targetPid, SIGKILL) == 0){
                        cout<<"Process "<< targetPid << " terminated successfully\n";
                }
                else{
                        perror("Failed to kill the process");
                }
        }
        return 0;
}
```

Day 4 Output

```
student@D001-38:~/Documents/capstone-3$ g++ --std=c++17 task4.cpp -o task4
student@D001-38:~/Documents/capstone-3$ ./task4
PID     CPU%    Memory (kB)     Name
1615    3.07015%        140727110998360%        (gnome-shell)%
5939    2.64061%        140727110998360%        (chrome)%
1486    0.989632%       140727110998360%        (Xorg)%
4148    0.958026%       140727110998360%        (nautilus)%
1050    0.914171%       140727110998360%        (mysqld)%
5721    0.381869%       140727110998360%        (kworker/10:1-events)%
4076    0.351272%       140727110998360%        (gedit)%
3795    0.227279%       140727110998360%        (kworker/10:0-events)%
4170    0.123096%       140727110998360%        (gnome-terminal-)%
76      0.0478193%      140727110998360%        (ksoftirqd/10)%
Enter PID to kill: 5939
Process 5939 terminated successfully
student@D001-38:~/Documents/capstone-3$
```

Day 5: Implement real-time update feature to refresh data every few seconds.

```
#include <iostream>
#include <filesystem>
#include <fstream>
#include <sstream>
#include <algorithm>
#include <vector>
#include <dirent.h>
#include <unistd.h>
#include <csignal>
#include <thread>
#include <chrono>
using namespace std;
namespace fs=std::filesystem;
struct ProcessInfo{
        int pid;
        string name;
        double cpuUsage;
        long memoryUsage;
};
string readFileValue(const string &path){
        ifstream file(path);
        string value;
        if (file.is_open()){
                getline(file,value);
```

```
        }
        return value;
}
double getSystemUptime(){
        ifstream file("/proc/uptime");
        double uptime;
        if (file.is_open()){
                file >> uptime;
        }
        return uptime;
}
ProcessInfo getProcessInfo(int pid,double systemUptime){
        ProcessInfo pinfo;
    pinfo.pid = pid;
        ifstream file("/proc/" + to_string(pid) + "/stat");
        string line;
        long utime=0, stime=0, starttime=0;
        if(file.is_open()){
                getline(file,line);
                istringstream ss(line);
                string token;
                int count=0;

                while(ss >> token){
                        count++;
```

```
                if(count==2) pinfo.name=token;
                else if(count==14) utime=stol(token);
                else if(count==15) stime=stol(token);
                else if(count==22) starttime=stol(token);
            }
        }

        ifstream memFile("/proc/" + to_string(pid) + "/status");
        if (memFile.is_open()){
            string key, value,unit;
            while(memFile >> key >> value >> unit){
                if (key=="VmRSS"){
                    pinfo.memoryUsage=stol(value);
                    break;
                }
            }
        }
        long total_time = utime + stime;
        double seconds = systemUptime - (starttime / sysconf(_SC_CLK_TCK));
        pinfo.cpuUsage = (total_time / sysconf(_SC_CLK_TCK)) / seconds * 100;
        return pinfo;


}
```

```
vector <ProcessInfo> getAllProcesses(){
	vector <ProcessInfo> processes;
	double systemUptime= getSystemUptime();
	for (const auto &entry : fs::directory_iterator("/proc")){
		if (entry.is_directory()){
			string filename=entry.path().filename().string();
			if (all_of(filename.begin(),filename.end(), ::isdigit)){
				int pid=stoi(filename);

processes.push_back(getProcessInfo(pid,systemUptime));
			}
		}
	}
	return processes;
}

void sortProcesses(vector<ProcessInfo> &processes, bool sortByCpu){
	if(sortByCpu){
		sort(processes.begin(),processes.end(),[](const ProcessInfo &a, const
ProcessInfo &b)
		{
			return a.cpuUsage > b.cpuUsage;
		});
	}
	else{
```

```
        sort(processes.begin(),processes.end(),[](const ProcessInfo &a, const
ProcessInfo &b)
        {
            return a.memoryUsage > b.memoryUsage;
        });
    }
}


int main(){
    char input;
    while (true){
        system("clear");
        vector <ProcessInfo> processes = getAllProcesses();
        sortProcesses(processes,true);
        cout<<"Press 'q' to quit";
        cout<<"PID\tCPU%\tMemory (kB)\tName\n";
        for (size_t i=0;i<min(processes.size(),size_t(10));i++){

cout<<processes[i].pid<<"\t"<<processes[i].cpuUsage<<"%\t"<<processes[i].memory
Usage<<"%\t"<<processes[i].name<<"%\n";
        }
        int targetPid;
        cout<<"Enter PID to kill: ";
        cin>>targetPid;
        if (targetPid == 0){
```

```
                        //continue refresh
                }
                else if(targetPid > 0){
                        if (kill(targetPid, SIGKILL) == 0){
                                cout<<"Process "<< targetPid << " terminated
successfully\n";
                        }
                        else{
                                perror("Failed to kill the process");
                        }
                }

                cout << "\nPress 'q' to quit or Enter to refresh: ";
                cin.ignore( );
                input = getchar( );
                if (input == 'q' || input =='Q')
                break;
                this_thread::sleep_for(chrono::seconds(2));
        }
        return 0;
}
```

Day 5 Output (Killing Process)

```
Press 'q' to quitPID    CPU%    Memory (kB)     Name
6696    3.79939%        140737058960760%        (chrome)%
1615    3.13593%        140737058960760%        (gnome-shell)%
1486    1.02253%        140737058960760%        (Xorg)%
4148    1.01235%        140737058960760%        (nautilus)%
1050    0.911457%       140737058960760%        (mysqld)%
6586    0.845166%       140737058960760%        (gnome-terminal-)%
4076    0.331241%       140737058960760%        (gedit)%
3795    0.21504%        140737058960760%        (kworker/10:0-events)%
5721    0.186109%       140737058960760%        (kworker/10:1-events)%
76      0.0471295%      140737058960760%        (ksoftirqd/10)%
Enter PID to kill: 6696
Process 6696 terminated successfully

Press 'q' to quit or Enter to refresh:
```

Day 5 Output (After Refreshing)

```
Press 'q' to quitPID    CPU%    Memory (kB)     Name
1615    3.17135%        140737058960416%        (gnome-shell)%
4148    1.12394%        140737058960416%        (nautilus)%
1486    1.04837%        140737058960416%        (Xorg)%
1050    0.911807%       140737058960416%        (mysqld)%
6586    0.453206%       140737058960416%        (gnome-terminal-)%
4076    0.344642%       140737058960416%        (gedit)%
3795    0.210823%       140737058960416%        (kworker/10:0-events)%
5721    0.156335%       140737058960416%        (kworker/10:1-events)%
76      0.052087%       140737058960416%        (ksoftirqd/10)%
5182    0.0364611%      140737058960416%        (kworker/u24:1-events_unbound)%
Enter PID to kill:
```